

Smart Decentralized Greenhouse – Hardware & Wiring Documentation

Phase 1 – Wiring and Hardware Bring-up

Controller: ESP32-WROOM-32E

Document Scope: Complete electrical specification from the 220-230 V mains source through every ESP32 Pin.

Control Logic: MQTT, and server layers are out of scope for this document.

Document Version	1.0
Date	June 19, 2026
Phase	1 out of 9 (hardware bring-up)
Status	Wiring locked – Ready to build

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1. Introduction

This document presents the complete hardware wiring and electrical architecture of the *Smart Decentralized Greenhous*, an *academic proof-of-concept prototype* intended for eventual production deployment.

The system is built around an **ESP32 microcontroller**, powered by a regulated **5 V DC power supply**, and integrates multiple analog and digital sensors, actuators, and control interfaces.

All subsystems have been designed to ensure stable operation, electrical safety, and logical pin allocation for efficient firmware development.

The build is organized as nine dependency-ordered phases.

Phase 1 – the subject of this document – wires every sensor and actuator to the controller and verifies that each one reads or switches correctly, with no control logic present.

Establishing a clean, verified hardware baseline here removes electrical faults as a variable for every later phase (threshold control, fuzzy arbitration, RBAC, vision, ledger).

The controlling design constraints, all reflected in the pin assignments that follow, are:

- ADC2 is unusable while Wi-Fi is active, so every analog sensor is placed on ADC1 channel (GPIO 32-39).
- Strapping Pins (GPIO 0, 2, 5, 12, 15) are avoided for driven loads, with one deliberate, justified exception (GPIO 5).
- Input-Only Pins (GPIO 34, 35, 36, 39) carry only analog inputs, never outputs.
- All actuators are binary (on/off), switched through an active-LOW relay module, with the 220 V grow-light circuit requiring guaranteed-off behaviour through boot.

2. System Overview

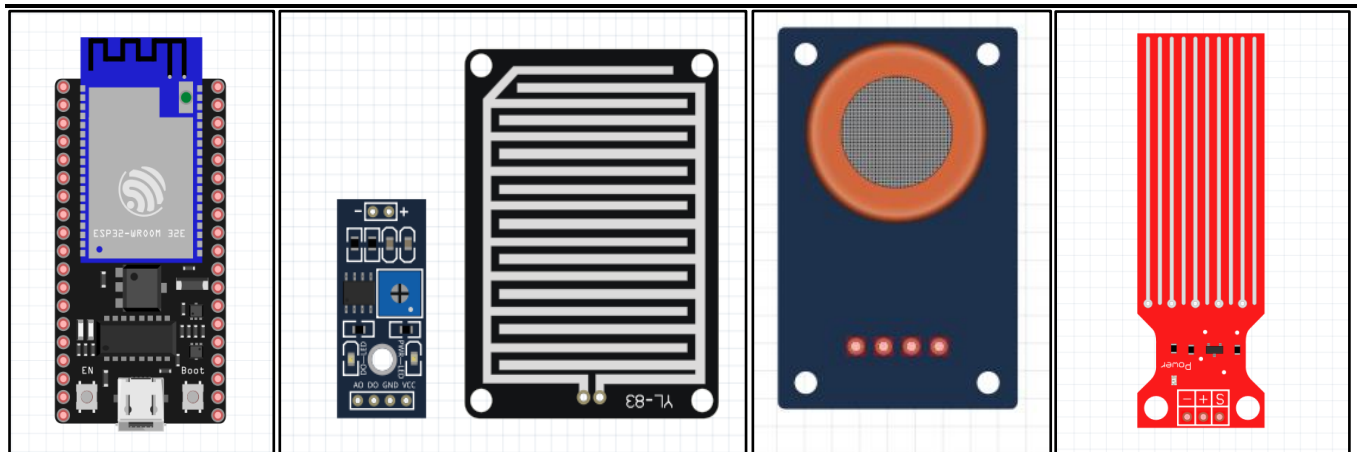
The full system splits at MQTT into an edge tier (this controller, must run standalone) and a server tier (Dokploy-hosted)

This section lists all hardware items used in the Smart CEA prototype, including their electrical type, voltage class, and primary function within the system.

NO.	Component	Quantity	Supply	Signal Type	Function
1	ESP32 Wroom-32E	1	5v	Logic 3.3v	Main Micro-Controller
2	Rain Sensor Module (MH-RD)	1	3.3v	AO	Detects rain or surface moisture on roof area
3	Gas Sensor (MQ-135)	1	5v	AO Via Volt Divider	Detects air quality and gas concentration
4	Water Level Sensor	2	3.3v	Analog	Detects soil or reservoir moisture level
5	LDR (Photo-Resistor)	2	3.3v	Analog	Measure ambient light sensitivity
6	Temperature-Humidity Sensor (DHT11 3-pins)	2	3.3v	Digital	Measures temperature and humidity (inside / outside)
7	BMP280 (Pressure-Temp Sensor)	2	3.3v	I ² C (SDA/SCL)	Reads pressure, temperature, altitude (inside / outside)
8	MG90S Micro Servo (180° mini servo)	2	5v	PWM	Controls side window mechanisms (in parallel)

9	MG996R High-Torque Servo	1	5v	PWM	Controls roof opening mechanism
10	DC Brushless Fans (5v 0.08A)	2	5v	Switched Via Relay	Airflow regulation (north / south sides)
11	Submersible mini Water Pump (DC)	1	5v	Switched Via Relay	Irrigation system
12	LED (Discreted with 220 ohm resistors)	10	5v	Switched Via Relay	Lighting (storage, green area, control room)
13	I ² C LCD Display 16x2	1	3.3v	I ² C (SDA/SCL)	
14	Rotary Encoder (HW-040)	1	5v	Digital Control Input	User interface for parameter adjustment
15	8-Channel Relay Module	1	5	Digital Control Input	Switches Fans, Pump, light
16	Power Supply Unit (S-25-5)	1	Alternative Current	-	Provides 5v DC power to all loads

Components

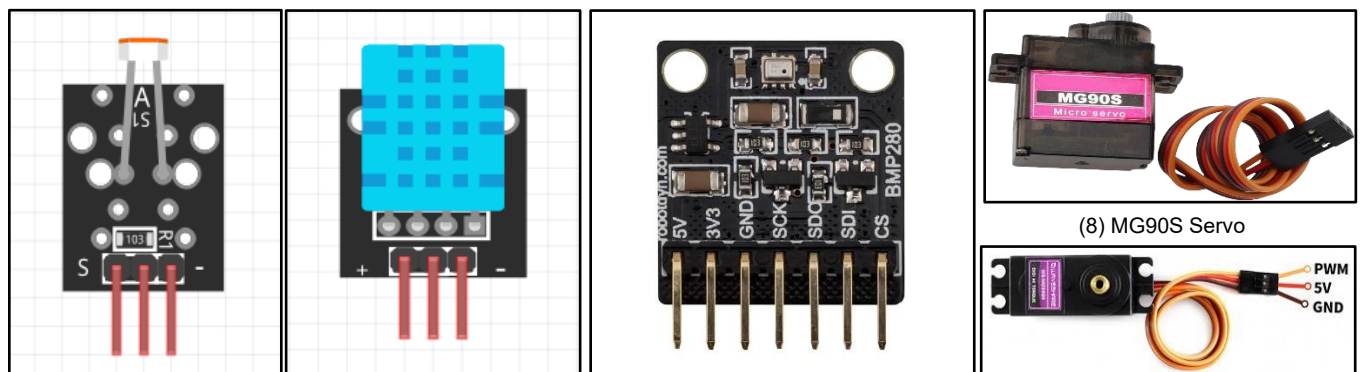


(1) ESP32-WROOM-32E

(2) Rain Sensor (MH-RD)

(3) Gas Sensor (MQ135)

(4) Water Sensor

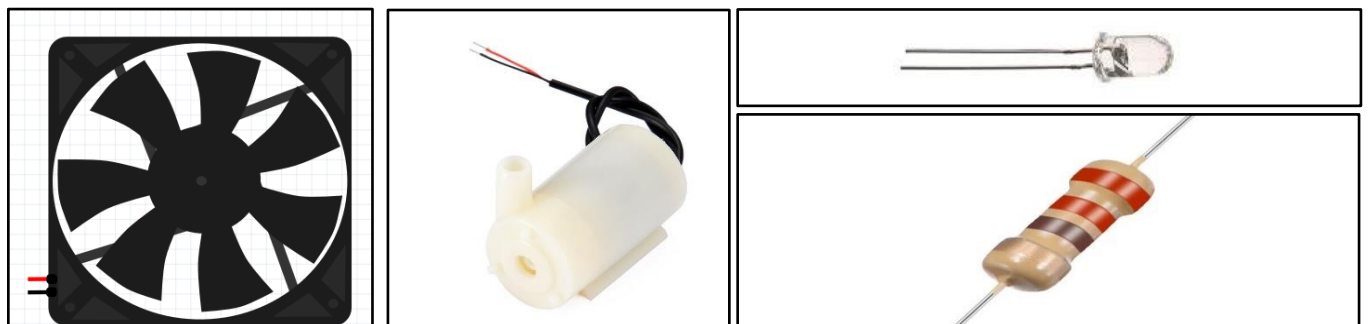


(5) LDR Photoresistor

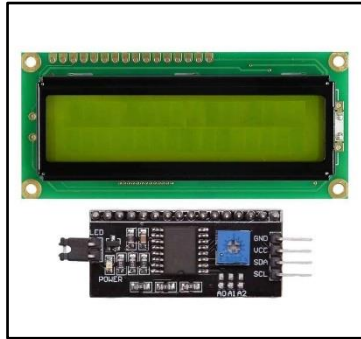
(6) DHT11

(7) BMP280

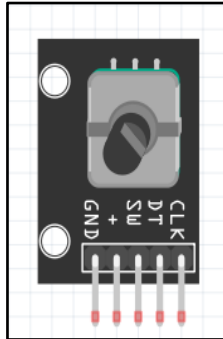
(9) MG996R Servo



(10) DC Brushless Fans 5v 0.08A

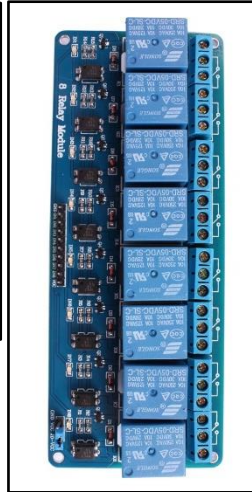
(13) I²C LCD Display 16x2

(11) Water Pump DC

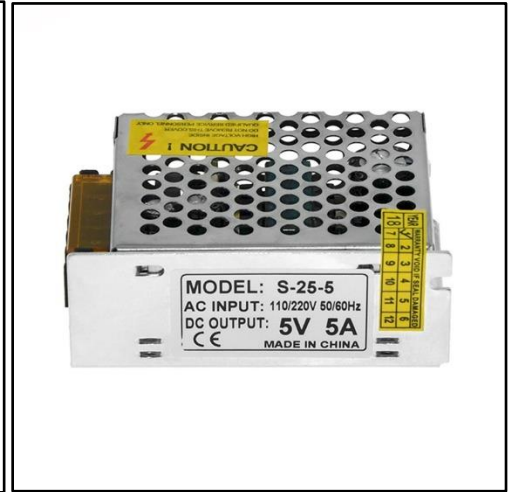


(14) Rotary Encoder

(12) LEDs & 220-ohm Resistors



(15) 8-Relay-Module



(16) Power Supply S-25-5

3. Electrical Color Coding

To maintain consistency and safety across all modules, the following color code standard is used throughout the circuit:

Color	Signal	Voltage Level
Red	+5v	5v DC
Yellow	+3.3v	3.3v DC
Black	GND	-
Yellow	SDA	3.3v logic
Blue	SCL	3.3v logic
Green	Analog Signal	0-3.3v
White	Digital Signal	3.3v logic
Purple	PWM – Control Output	3.3v logic
Brown	Configuration	-

4. Power Supply Architecture

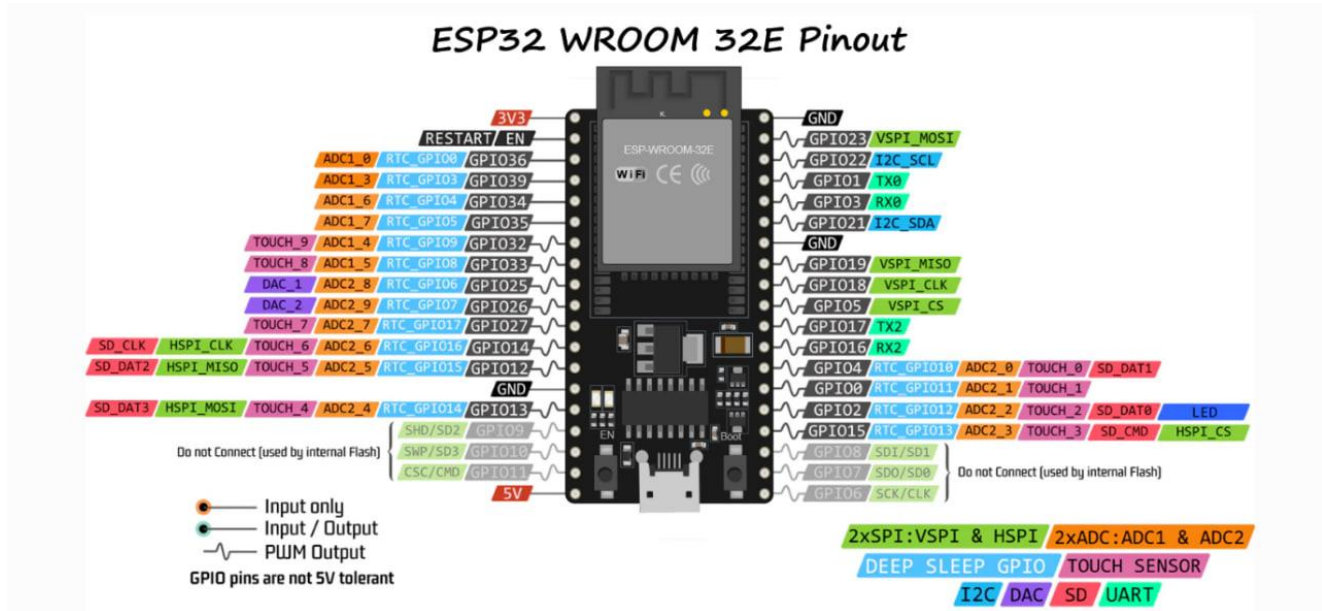
— Primary Power Source

- **Power Supply Unit (PSU) – 5v DC**
- **Distribution:**
 - **5v Rail:** Relays – Servos – Fans – Pump – MQ135 Sensor – Led Groups – ESP32
 - **3.3v Rail:** Sensors – Logic Interfaces –
 - **Ground:** All modules connected to a common reference ground

— ESP32 Power Input

- PSU +5v → ESP32 5v (Vin Pin)
- PSU GND → ESP32 GND
- Onboard 3.3v regulator supplies logic modules

5. ESP32 Pin Architecture



Pin Type	Range	Description
ADC1	GPIO32-39	Analog Inputs
PWM Digital I/O	GPIO4-19, 21-23, 25-27	Outputs & PWM Control
I ² C Default	GPIO21 (SDA) , GPIO22 (SCL)	Shared Bus for LCD and BMP280 Sensors
Boot Sensitive Pins	GPIO0 , GPIO2 , GPIO15	Inputs with pull-ups
Restricted Pins	GPIO6-11	Flash Memory Bus – NOT USED!
UART Pins	GPIO1 (TX0) , GPIO3 (RX0)	Programming & Debugging Purposes

6. Sensor Wiring Configuration

— Analog Sensors

Sensor	EDP32 Pin	Signal Type	Supply	Notes
Rain Sensor	GPIO36	Analog	3.3v	DO unused
MQ-135 Gas Sensor	GPIO39	Analog	5v	AO via divider
Water Sensor #1	GPIO34	Analog	3.3v	-
Water Sensor #2	GPIO35	Analog	3.3v	-
LDR #1 (INNER)	GPIO32	Analog	3.3v	-
LDR #2 (OUTER)	GPIO33	Analog	3.3v	-

— Digital Sensor

Sensor	EDP32 Pin	Signal Type	Supply	Notes
DHT11 #1 (INNER)	GPIO25	Analog	3.3v	10(k-ohm) pull up if not built in
DHT11 #2 (OUTER)	GPIO26	Analog	3.3v	10(k-ohm) pull up if not built in

— I²C Sensors & Display

Sensor	SDA	SCL	Supply	Notes
LCD (I ² C)	GPIO21	GPIO22	5v	Address: 0x27 or 0x3F
BMP280 #1 (INNER)	GPIO21	GPIO22	3.3v	— CS → 3.3v — SDO → GND Address: 0x76
BMP280 #2 (OUTER)	GPIO21	GPIO22	3.3v	— CS → 3.3v — SDO → GND Address: 0x77

7. Actuators and Outputs

8. Relay Module (6 Channels Used Out Of 8 Channels 6/8)

Function	Relay Input	ESP32 Pin	Signal Type	Supply
South Fan	IN1	GPIO19	Digital Output	5v
Storage Lights	IN2	GPIO18	Digital Output	5v
C-Room Lights	IN3	GPIO5	Digital Output	5v
Green Area Lights	IN6	GPIO17	Digital Output	5v
Water Pump	IN7	GPIO16	Digital Output	5v
North Fan	IN8	GPIO4	Digital Output	5v

9. Servo Motors

Servo	ESP32 Pin	Signal	Supply	Notes
MG90S (Windows Pair) x2	GPIO27	PWM	5v PSU	Signal Shared In Parallel
MG996R (Roof)	GPIO23	PWM	5v PSU	Separated Control Line Signal

10. Rotary Encoder

Encoder Pin	ESP32 Pin	Mode	Supply	Description
VCC	-	3.3v	Power	-
GND	-	GND	Common Ground	-
SW (BUTTON)	GPIO0	Input-Pullup	3.3v Output via ESP32 Pin	Push Button Input

DT	GPIO2	Input-Pullup	3.3v Output via ESP32 Pin	Channel B
CLK	GPIO15	Input-Pullup	3.3v Output via ESP32 Pin	Channel A

11. Reserved Pins

Pin	Description	Status
GPIO6 → GPIO11	SPI Flash Interface	Reserved
GPIO12	Boot strap / Flash Voltage	Unused
GPIO1 (TX0) , GPIO3 (RX0)	USB-UART Interface	Reserved for Programming & Debugging
Any un-assigned GPIO	Future Expansions	FREE